Customer Shopping Behavior Analysis

Comprehensive Data Analysis Report

Dataset Size	Analysis Period	Report Date
3,900 Transactions	**Multiple Seasons**	**November 30, 2025**

Analysis Tools:

Python (Pandas, NumPy) | PostgreSQL (pgAdmin) | Power BI

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across multiple product categories. The goal is to understand:

- Who the customers are (age, gender, subscription status, location)
- How they shop (spend amount, discounts, shipping type, repeat purchases)
- Which products and segments drive the most revenue

The final deliverables are:

- Cleaned dataset in Python
- Business queries answered in PostgreSQL (pgAdmin)
- An interactive Customer Behavior Dashboard in Power BI

2. Dataset Summary

Dataset Overview

Metric	Value
Rows	3,900
Columns	18

Key Features

The dataset includes the following key features:

- **Customer demographics:** Age, Gender, Location, Subscription Status
- **Purchase details:** Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
- **Shopping behavior:** Discount Applied, Promo Code Used, Previous Purchases, Payment Method, Frequency of Purchases, Review Rating, Shipping Type

Summary Statistics - Main Features

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Age	3,900	43.89	15.17	18	29	43	58	70
Purchase Amt ($)	3,900	59.76	26.96	20	36	60	80	100
Prev Purchases	3,900	24.84	14.82	1	13	25	37	50
Review Rating	3,863	3.75	0.69	2.5	3.1	3.7	4.4	5.0

Summary Statistics - Additional Features

Feature	Count	Unique	Top	Freq
Gender	3,900	2	Male	2,209
Category	3,900	4	Clothing	1,018
Location	3,900	44	Montana	163

Feature	Count	Unique	Top	Freq
Season	3,900	4	Spring	1,003
Shipping Type	3,900	2	Free Shipping	1,964
Subscription	3,900	2	No	2,640
Payment Method	3,900	4	Credit Card	1,650
Frequency	3,900	3	Fortnightly	1,366

Missing data: 37 missing values in the Review Rating column. These were filled using the median rating within each product category to avoid the impact of outliers.

3. Exploratory Data Analysis Using Python

Using pandas in Jupyter Notebook, comprehensive data preprocessing was performed:

- **Data loading:** Imported the CSV into a pandas DataFrame for analysis.
- **Initial exploration:** Used df.info() and df.describe() to understand data types, ranges, and summary statistics.
- **Handling missing values:** Checked nulls with df.isnull().sum(). Filled 37 missing values in Review Rating using the median rating within each product category.
- **Column standardization:** Renamed columns to snake_case (e.g., Customer ID -> customer_id).
- **Feature engineering:** Created an age_group column by binning ages into: Young Adult, Adult, Middle-aged, Senior.
- **Database integration:** Loaded the cleaned DataFrame into PostgreSQL using SQLAlchemy for further SQL-based analysis.

4. Data Analysis Using SQL (Business Questions)

All queries were run in PostgreSQL (pgAdmin) on the cleaned table. The following business questions were answered through detailed SQL analysis.

Q1. What is the total revenue generated by male vs. female customers?

Result:

Metric	Female	Male	Ratio
Customers	1,691 (43%)	2,209 (57%)	1.3x
Total Revenue	$75,191	$157,890	2.1x
Avg Purchase/Customer	$44.47	$71.48	1.6x

KEY INSIGHT: Male customers generate 2.1x more total revenue than females, driven by two compounding factors: they represent 57% of the customer base AND their average purchase per customer is 1.6x higher ($71.48 vs $44.47). Both dimensions — acquiring more female customers and increasing female average order value — present untapped upside. The $27 per-customer gap warrants investigation into product assortment, pricing, and marketing resonance by gender.

Q2. Which customers used a discount but still spent more than the average purchase amount?

Result:

Identified 839 such orders where customers used a discount yet spent above the global average purchase amount ($59.76).

KEY INSIGHT: High-value customers respond to discounts but still spend more than average. Targeting them with personalized offers can increase revenue without hurting margins too much.

Q3. Which are the top 5 products with the highest average review rating?

Result:

Top 5 by average rating: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)

KEY INSIGHT: These products are good candidates for being highlighted in marketing campaigns, bundles, and recommendation sections due to strong customer satisfaction.

Q4. Compare the average purchase amounts between Standard and Express shipping.

Result:

Standard shipping: $58.46 | Express shipping: $60.48

KEY INSIGHT: Customers choosing Express shipping spend slightly more on average, indicating higher purchase urgency or higher-value baskets.

Q5. Do subscribed customers spend more? Compare average spend and total revenue.

Result:

Average spend is very similar between subscribed and non-subscribed customers. Total revenue from non-subscribers is much higher due to their larger count.

KEY INSIGHT: There is still room to convert heavy non-subscriber buyers into subscribers with targeted benefits.

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

Result:

Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), Pants (47.37%)

KEY INSIGHT: These products rely heavily on discounts to sell. Pricing and promotion strategy should be reviewed to improve profitability.

Q7. Segment customers into New, Returning, and Loyal based on total previous purchases.

Result:

Loyal: 3,116 customers | New: 83 customers | Returning: 701 customers

KEY INSIGHT: The majority of customers are Loyal, indicating strong retention. However, the New segment is small (only 83), suggesting that acquiring completely new customers may be an area to focus on.

Q8. What are the top 3 most purchased products within each category?

Result:

Accessories: Jewelry (171), Sunglasses (161), Belt (161)

Clothing: Blouse (171), Pants (171), Shirt (169)

Footwear: Sandals (160), Shoes (150), Sneakers (145)

Outerwear: Jacket (163), Coat (161)

KEY INSIGHT: These products are the core catalog items and should be kept in stock, prominently featured, and considered for bundles.

Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe

Result:

Subscription Status - No: 2,518 repeat buyers | Yes: 958 repeat buyers

KEY INSIGHT: Many repeat buyers still do not subscribe, which is a missed opportunity. There is a strong candidate group (2,518 repeat buyers without subscription) for subscription-focused campaigns.

Q10. What is the revenue contribution of each age group?

Result:

Young Adult: $62,143 | Middle-aged: $59,197 | Adult: $55,978 | Senior: $55,763

KEY INSIGHT: All age groups generate similar revenue, but Young Adults contribute slightly more. Marketing can be balanced but with a slight focus on younger users, especially for digital channels.

5. Interactive Dashboard in Power BI

An interactive Customer Behavior Dashboard was built in Power BI to present the insights visually. The dashboard includes:

KPIs

- Number of customers: ~3.9K
- Average purchase amount: $59.76
- Average review rating: 3.75

Visualizations

- % of customers by subscription status
- Revenue by category
- Sales by category
- Revenue by age group
- Sales by age group

Interactive Slicers

- Subscription Status
- Gender
- Category
- Shipping Type

6. Business Recommendations

Based on the comprehensive analysis, the following key recommendations are proposed:

1. Increase Subscription Conversions

Many heavy buyers are not subscribed. Offer targeted incentives (exclusive discounts, early access, loyalty rewards) to convert them. Focus on the 2,518 repeat buyers without subscriptions as high-priority targets.

2. Optimize Discount Strategy

Products like Hat, Sneakers, Coat, Sweater, and Pants are highly discount-dependent. Test reduced discount levels or value bundles to protect margins while keeping volume. Consider dynamic pricing strategies.

3. Leverage Top-Rated and Best-Selling Products

Highlight Gloves, Sandals, Boots, Hat, and Skirt in marketing campaigns, homepages, and recommendation engines. These products have the highest customer satisfaction and should be featured prominently.

4. Focus on High-Value Segments

Young Adult and Middle-aged groups drive the most revenue. Design age-specific promotions for these segments. Develop targeted digital campaigns for younger demographics.

5. Strengthen New Customer Acquisition

The New segment is very small (83 customers). Invest in awareness/acquisition campaigns and then use the loyalty engine to move them into Returning and Loyal segments. Develop referral programs leveraging loyal customers.

6. Promote Express Shipping for Upsell

Express shipping customers spend slightly more. Encourage upgrades to Express with small fees or subscription perks. Consider free express shipping thresholds to increase basket sizes.

7. Gender-Targeted Marketing

Male customers generate 2.1x more total revenue — driven by both higher representation (57% of base) and 1.6x higher average spend per customer ($71.48 vs $44.47). While maintaining female-focused initiatives, consider expanding male-oriented product lines and marketing channels, and investigate what drives the $27 per-customer gap to identify cross-sell and upsell opportunities for female shoppers.

8. Product Bundle Strategy

Create strategic bundles using top-performing products from different categories (e.g., Jewelry + Blouse + Sandals) to increase average order value and cross-category purchases.

7. Conclusion

This comprehensive analysis of 3,900 customer transactions has revealed critical insights into shopping behavior, customer segmentation, and revenue drivers.

Key Findings

- A strong loyal customer base (3,116 customers) provides a solid foundation for growth
- Significant opportunity exists in converting 2,518 repeat buyers to subscribers
- Product performance varies significantly, with clear winners that should be leveraged
- All age groups contribute meaningfully to revenue, enabling diversified marketing strategies
- Discount dependency for certain products presents both a risk and an optimization opportunity

Next Steps

- Implement A/B testing for subscription conversion campaigns

- Review and optimize pricing strategy for discount-heavy products
- Develop targeted acquisition campaigns to grow the new customer segment
- Create personalized marketing journeys based on customer segments
- Monitor KPIs through the Power BI dashboard for continuous improvement

The combination of Python data preprocessing, SQL-based business intelligence, and Power BI visualization provides a robust framework for data-driven decision making. Regular updates to this analysis will ensure strategies remain aligned with evolving customer behaviors and market conditions.

Generated with Python (fpdf2) | Customer Shopping Behavior Analysis Report